

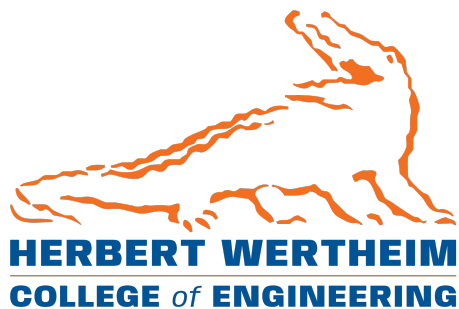
T1: Alpha Test Plan

EEL4908 CpE Design 2

Hardened RISC-V Exam Computer

September 9, 2026

Cannon Spencer George Rushevich Justin Reyes Gabriel Velazquez



1 Scope

This plan is for the alpha build. The pass criterion is **Safe Exam Browser running on the CI-built Debian image** on the Orange Pi RV.

In scope. Image flash and boot; network; build and launch of `seb-linux` on that Debian; basic lockdown once SEB is up.

Out of scope. Instructor control plane (start/stop jobs, fleet UI), Canvas exam completion, and kernel hardening.

Lab baseline. Default login `orange_pi/orange_pi`. UART 115200. *TODO*

2 Expected behavior

TODO

2.1 Board and image

CI in `reptilian-riscv` publishes a Debian `os.img.xz`. `flash-os.sh` writes it to SD. After power, the board reaches a Linux login (UART and SSH). Wi-Fi or Ethernet yields IPv4 and outbound connectivity.

2.2 SEB on Debian

- `seb-linux` builds on riscv64 using the WebKitGTK fallback (no Qt6 WebEngine package).
- SEB launches on the Debian graphical session.
- While running: fullscreen; ordinary desktop exit paths do not dismiss the kiosk; policy-blacklisted processes are refused or terminated.

TODO

2.3 seb-agent heartbeat

`install-software.sh` copies `seb-agent` to the board. The agent POSTs a JSON heartbeat to a control URL every 5 s.

3 Test procedures

Each test: preconditions, steps, pass/fail.

3.1 Bring-up

ID	Procedure
HW-01	Flash and boot. <code>diskutil list</code> (never <code>disk0</code>); <code>./scripts/flash-os.sh -device /dev/rdiskN</code> ; power with UART; <code>screen /dev/cu.usbserial-* 115200</code> . Pass: Linux login, not an <code>initramfs</code> emergency shell.
HW-02	SSH. Same LAN; <code>ssh orangepi@<board-ip></code> . Pass: session opens; <code>uname -m</code> is <code>riscv64</code> .
HW-03	Outbound network. <code>ping -c 4</code> to a known host (or campus DNS). Pass: replies, or a documented firewall exception.

Table 1: Bring-up tests.

3.2 SEB

ID	Procedure
SW-01	Build seb-linux on the board. Pre: HW-02; Qt6 base + WebKitGTK deps. Steps: <i>TODO</i> Pass: a SEB binary exists. If BUG-01/02 recur, log them and fail this test.
INT-01	SEB launches. Pre: SW-01; graphical session. Steps: <i>TODO</i> Pass: SEB window/kiosk is visible on the board display.
INT-02	Lockdown smoke. Pre: INT-01. Steps: try standard exit shortcuts; optionally start a blacklisted app if policy defines one. Pass: cannot leave by those paths; <code>refuse-to-start</code> or <code>terminate</code> matches policy.

Table 2: SEB tests.

3.3 seb-agent heartbeat

ID	Procedure
SW-02	seb-agent heartbeat. Pre: HW-03; laptop HTTP server on port 8000; <code>install-software.sh</code> with that <code>-control-url</code> . Run <code>~/seb-agent</code> on the board. Pass: <code>POST /heartbeat</code> from the board about every 5 s.

Table 3: seb-agent heartbeat.

4 Traceability

Expected behavior	Tests
Debian flash/boot/network (§2.1)	HW-01, HW-02, HW-03
SEB builds, launches, lockdown (§2.2)	SW-01, INT-01, INT-02
seb-agent heartbeat (§2.3)	SW-02

Table 4: Expected behavior to tests.